

Cloud DMZ Perimeter Design Proposal

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Reviewers and Contributors

Quick Links

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Name	Group	Role
Wynston, Michael (US - Florida)	PE	Sponsor
Kirichenko, Anton (Remote - AUS New South Wales)	PE	Contributor
Peters, Scott (Omaha)	PE	Contributor
Butau, Shaima (Berkeley Heights Remote)	PE	Contributor
Onaiwu, Etinosa (Columbus)	PE	Contributor
Ali, Wajid (Columbus)	PE	Contributor
Uchendu, Onyekachi (Columbus)	PE	Contributor

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https://aws.amazon.com/blogs/networking-and-content-delivery/centralized-outbound-inspection-architecture-in-aws-cloud-wan/	Centralized Outbound Inspection
https://aws.amazon.com/blogs/networking-and-content-delivery/centralized-outbound-inspection-architecture-in-aws-cloud-wan/	Hybrid Inspection via Cloud WAN Service Insertion
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https://learn.microsoft.com/en-us/security/zero-trust/azure-networking-visibility	Zero Trust Networking Visibility
https://linkedin.com/pulse/inspecting-every-packet-gcp-landing-zone-gaurav-patel	Inspecting Every Packet in GCP Landing Zone

1.Introduction

The Demilitarized Zone (DMZ) is a foundational security construct in on-premises network architecture, an isolated network segment that sits between the untrusted public internet and the trusted internal corporate network. It hosts public-facing services (web servers, API gateways, reverse proxies, load balancers) while enforcing strict ingress /egress controls through firewalls and access control lists on both perimeters.

As enterprises migrate workloads to the cloud, the challenge is not simply lifting a DMZ into a virtual machine, it is re-architecting the security boundary model for a distributed, elastic, software-defined environment where the traditional notion of a hard perimeter dissolves.

This reference architecture defines the strategic direction and design patterns for establishing an equivalent cloud DMZ, one that preserves the security intent of the on-premises model while embracing cloud-native capabilities: micro-segmentation, identity-aware access control, managed inspection services , and Zero Trust principles.

Every cloud DMZ design decision documented herein is anchored to two foundational baselines: (1) the current on-premises secure DMZ architecture and (2) Global Cybersecurity Services (GCS) Ports and Services governance framework.

2.0 Proposed On-Premises secure DMZ Design- Sandwich Model

The following has been defined as the standard Fiserv go-forward on-premises design which is currently being finalized. This architecture applies uniformly to all externally facing traffic categories: internet traffic, client connectivity, and third-party/partner connectivity.

The established architecture is a classic sandwich model with the following layering:

- External (Perimeter) Firewall - faces the public internet; handles all inbound and outbound internet traffic. This is the outermost security enforcement boundary.
- DMZ Segment - sits between the two firewall layers; hosts all resources that must be reachable from untrusted networks (e.g., web servers, external DNS, VPN concentrators).
- Internal (Inside) Firewall - separates the DMZ from the trusted internal corporate network. No traffic reaches internal resources without traversing this second enforcement boundary.

Traffic Category	DMZ Sandwich Applies?	Notes
Internet Traffic	YES - mandatory	All inbound/outbound internet traffic traverses outer FW DMZ inner FW
Client Connectivity	YES - mandatory	Remote access / VPN concentrators reside in DMZ; no direct access to internal
Third-Party / Partner Traffic	YES - mandatory	Partner circuits terminate in DMZ; proxied to internal via inner FW policy

2.1 Proxy and Connectivity Requirements

A key design constraint confirmed by cyber governance is that systems requiring direct internet connectivity cannot be placed directly on the internal trusted network segment. This principle is non-negotiable and carries over in full to the cloud architecture:

- Any device or service that communicates directly with the internet (e.g., VPN devices, external DNS servers, web servers) must sit in the DMZ.
- A proxy or termination layer (load balancer or similar) is required to terminate and relay connectivity between the internet-facing and internal zones.
- This ensures no unmediated path exists between the internet and internal network resources.

3. Cloud DMZ- Conceptual Mapping

Cloud environments do not have physical firewalls, VLANs, or hardwired network topologies. Instead, they offer software-defined constructs that must be deliberately configured to mirror the security intent of the on-premises model. The mapping below establishes architectural equivalence and explicitly identifies which Cybersecurity governance requirement each cloud control satisfies.

O n- Pr e m i s e s C o n s t r u c t	Cloud Equivalent	Notes	Cyber securi ty Requi reme nt
VL A N / D M Z S u b n et	Inspection VPC within Cloud WAN (AWS) / Virtual WAN (Azure) / NCC (GCP) core network; isolated segment type in the WAN fabric with no direct peering to spoke VPCs / VNets	Network-layer isolation; segment is a first-class construct in the global WAN core, not an external VPC attachment	Section 2.0 Sandwich Model
O ut er Fi re w all	NGFW/Cloud Native firewall deployed within the Inspection Segment of Cloud WAN / Virtual WAN / NCC- all internet ingress is steered through the inspection segment before reaching any spoke. Inspection Segment egress firewall policy (within Cloud WAN / VWAN / NCC) + Security Groups on spoke VPCs/VNets	North-South & East-West enforcement boundary; firewall lives inside the Inspection Segment and applied at segment level	Section 2.0 / GCS Port Control / Sandwich Model

DMZ Segment	Inspection Segment in the Cloud WAN / Virtual WAN / NCC core network-mandatory transit for all north-south and east-west traffic across the backbone; spokes route through the inspection segment by WAN routing policy	WAN-native DMZ; inspection segment enforced at the routing layer	Section 2.0 / Section 2.1
Proxy / Termination Layer	L7 Managed Load Balancer with WAF deployed within the Inspection Segment, mandatory between internet and internal spokes NLB, EC2 EKS, Network firewall Proxy for inspection of VPC outbound connection	No direct path internet spoke; all connections terminate at LB in inspection segment. Network firewall proxy for inspection and filtering of VPC egress connection to prevent data exfiltration	Section 2.1 Proxy Requirement
Web Application Firewall	Cloud-managed WAF (managed + custom rules) attached to LB in Inspection Segment	HTTP inspection at WAN ingress tier	GCS Section 6 (insecure services)
IPS / IDS	NGFW with IPS signatures inline within Inspection Segment OR managed cloud IDS service	Signature + behavioral; applied at segment level	GCS Port Control Policy

B as ti o n / Ju m p H ost	Managed Bastion Service (no public IP, session-recorded) provisioned within a dedicated management segment	System Session Manager access to instances ; all IP-based admin paths traverse inspection segment	Access Control Requirement
A C Ls	Cloud WAN / VWAN / NCC segment routing policies + Security Groups + NACLs on spoke VPCs/VNets + NGFW/Cloud-native firewall policy in inspection segment	Layered L3/L4 control; segment policy is the outermost layer	GCS Section 3
N et Fl ow / P C AP	VPC/VNet Flow Logs + Cloud WAN / VWAN / NCC flow telemetry + NGFW logs Cloud SIEM (Splunk + Thousand eyes)	Centralized logging with visibility across all segments and attachments	GCS Logging Requirement
C er tif ic at e M a n ag e m e nt	Managed PKI / Certificate Manager integrated with LB and WAF in Inspection Segment	TLS offload, mTLS at WAN ingress tier	GCS Section 3 Data Protection

Malware Scanning	AWS Network Firewall & Azure Firewall does not support native malware scanning and requires a dedicated 3rd-party or managed scanning service deployed separately within the Inspection Segment. However services like AWS Guarduty provides malware protection for EC2 and S3	Mandatory for all inbound files; positioned before any spoke admission	GCS Section 9
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4.0 Defined Use Cases for Cloud-Based DMZ

The following enterprise use cases drive the need for a cloud-based DMZ. Each case is evaluated against the on-premises sandwich model design intent and the applicable GCS Ports and Services governance constraints.

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4.1 Public-Facing Web and API Services

- External-facing web applications (portals, self-service platforms) hosted in cloud but backed by on-premises or cloud-native databases or internal APIs.
- Integration with SaaS platforms where the public endpoint must be segmented from the trusted environment.

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4.2 Hybrid Cloud Connectivity and Shared Services

- Consolidated entry point for all ingress/egress traffic — internet ingress arrives at the Inspection Segment VPC; on-premises and VPN connectivity attaches to the Hybrid Segment in the Cloud WAN / Virtual WAN / NCC backbone.
- Shared services construct for cloud-based service endpoints — a reserved VPC where all spokes access centralized services rather than creating redundant private endpoints in individual VPCs.

4.3 B2B Partner and Third-Party Integration

- API-based partner integrations: partner systems calling enterprise APIs are terminated in an isolated segment; the internal application tier is never directly reachable from the partner network.
- B2B partner data exchange: Partner infrastructure integration for file-based or message-based data exchange.

4.4 Multi-Cloud and Cloud-to-Cloud Traffic Inspection

- Cross-cloud service calls: a workload in AWS calling a service in Azure should traverse the Inspection Segment in the respective Cloud WAN / Virtual WAN core network and not route directly over uncontrolled backbone peering. The Inspection Segment NGFW enforces policy on that east-west inter-cloud flow before it exits the WAN fabric.
- SaaS-to-cloud integration: Workday, ServiceNow, or similar SaaS platforms calling enterprise cloud APIs, the Inspection VPC with cloud-native firewall or 3rd-party NGFW inspects those calls before they reach internal cloud workloads.
- Cloud-to-cloud data pipeline security: data movement between cloud environments (ETL, replication) passes through an inspection VPC layer to detect data exfiltration or anomalous volume.

5.0 Reference Architecture

The recommended pattern is a global WAN backbone topology anchored in AWS Cloud WAN, Azure Virtual WAN, and GCP Network Connectivity Center (NCC) as the underlying network fabric, replacing traditional hub-and-spoke constructs such as Transit Gateway, VNet peering hubs, and VPC peering meshes. Each WAN platform natively supports named network segments (Cloud WAN Segments, Virtual WAN routing intents and tables, NCC routing groups). The design intent is to define a dedicated Inspection Segment within each core network to enforce centralized egress inspection per region, while allowing two distinct ingress inspection strategies: distributed (per-VPC) or centralized (per-region Inspection VPC).

Spoke VPCs and VNets attach to their respective WAN backbone and are steered through the Inspection Segment by routing policy — not by static VPC peering or manually maintained route tables. The Inspection Segment hosts all firewall capability: the NGFW (cloud-native or third-party via GWLB or ILB), L7 Application Load Balancers, and WAF. The firewall enforces both external (internet-facing) and internal (spoke-facing) policy through separate ingress and egress rule sets, consolidating what the on-premises model expressed as a sandwich of two discrete firewall tiers into a single, auditable enforcement point. The WAN backbone guarantees that every spoke joining the network is automatically subject to Inspection Segment policy — with no spoke permitted to bypass inspection.

5.1 Traffic Flow and inspection Design Principles

The traffic flow principles below govern all cloud DMZ routing within the Cloud WAN backbone. Because the two supported inspection patterns handle inbound internet traffic differently, the principles are organized into two groups: common principles that apply to both patterns, followed by pattern-specific traffic flows for Distributed Ingress / Centralized Egress and Centralized Ingress & Egress respectively.



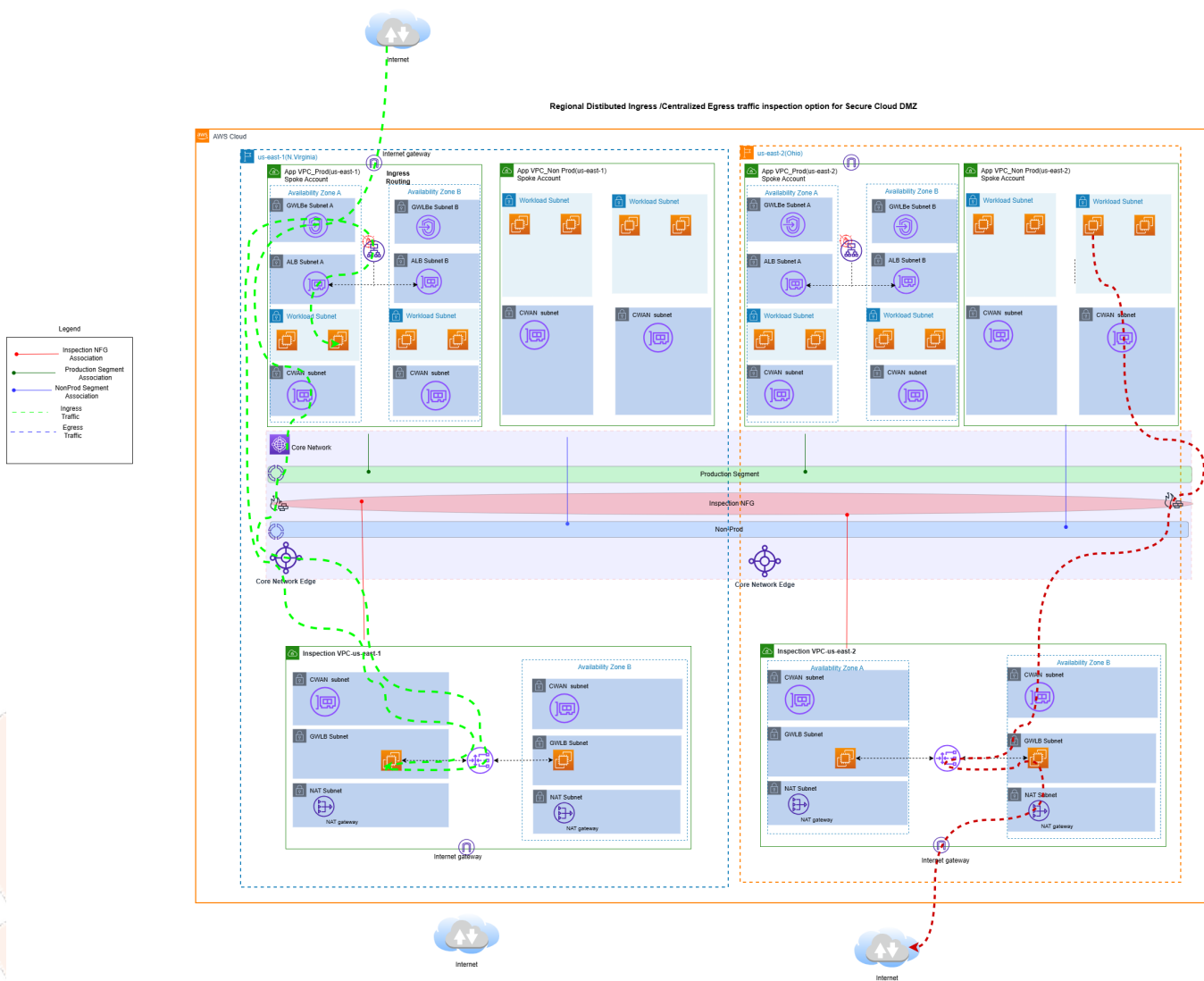
5.1.1 Common Traffic Flow(Internet Egress) Principles (Both Patterns)

The following principles are invariant across both inspection architectures and are enforced by Cloud WAN segment routing policy at the core network layer:

- All outbound internet traffic from spoke VPCs is routed through the WAN backbone to the regional Inspection VPC NGFW for policy enforcement before exiting to the internet. No spoke VPC holds a direct Internet Gateway.
- East-west traffic between spoke VPCs traverses the Cloud WAN backbone and is steered through the Inspection Segment. Segment routing policy prevents any direct spoke-to-spoke path — all inter-spoke traffic is subject to the same NGFW enforcement as north-south traffic.
- Each region maintains its own independent Inspection VPC and Inspection Segment, eliminating cross-region hairpinning for intra-region flows.
- Hybrid connectivity (Direct Connect, ExpressRoute, Interconnect, S2S VPN) attaches to a dedicated Hybrid Segment. On-premises traffic is routed through the Inspection Segment NGFW before reaching any spoke VPC, mirroring the inner firewall boundary of the on-premises model.
- Partner and M&A connectivity attaches via a dedicated M&A Segment, isolated from both the Hybrid Segment and Production Segment. All partner-originated traffic will not traverse the Inspection Segment NGFW before reaching any spoke because inspection has already been carried out by the Palo Alto firewalls in the Fiserv-M&A boundary
- Cross-cloud traffic (AWS to Azure, AWS to GCP) exits one WAN backbone through its Inspection Segment, traverses the inter-cloud link, and enters the destination WAN backbone through its own Inspection Segment — ensuring dual inspection on all cross-cloud flows

5.1.2 Traffic Flows — Distributed Ingress / Centralized Egress

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In this pattern, Gateway Load Balancer endpoints are deployed in each workload VPC, which requires its own Internet Gateway and per-AZ NGFW endpoint in a dedicated firewall subnet. Inbound internet traffic is received and, leveraging ingress routing, is routed to the Gateway Load Balancer endpoint. The endpoint forwards the traffic to the regional Inspection VPC, where the firewalls perform inspection before returning the traffic to the local Application Load Balancer in the spoke account. Outbound traffic from all spoke VPCs is centralized through a shared regional Inspection VPC. In this architecture, the Cloud WAN core network enforces two named segments: the Production segment and Non-Production segment. Both segments route through the Inspection segment via send-via policy, as do all other segments.

Inbound Internet Traffic (Distributed Ingress):

- An inbound internet request is received at the **workload VPC Internet Gateway** in the serving region (e.g., us-east-1 or us-east-2). Each workload VPC has its own IGW — there is no shared inbound entry point across workload VPCs.

- The IGW ingress route table (ingress routing) directs the packet to the **Gateway Load Balancer endpoint** in the Firewall Subnet of the same AZ. This is enforced by an edge association on the IGW route table via ingress routing, ensuring traffic cannot reach the ALB before passing through the firewall.
- The Gateway Load Balancer endpoint forwards the traffic to the **regional Inspection VPC**, where the NGFW applies stateful L4/L7 inspection, IPS, threat prevention, and URL filtering policy. Traffic that does not match an allow rule is dropped and logged
- Approved traffic is **returned from the Inspection VPC** to the **local ALB in the workload VPC**. The WAF evaluates the HTTP/S request against OWASP managed rule groups and any custom rule sets (e.g., rate limiting, geo-blocking, SQLi/XSS prevention).
- The ALB terminates TLS and load-balances approved requests to the respective backend EC2 instances, Fargate ECS tasks, or other compute targets in the **workload VPC**

Outbound Internet Traffic (Centralized Egress):

- A spoke VPC (VPC_Prod) initiates an outbound internet request. The subnet route table directs 0.0.0.0/0 to the Cloud WAN Attachment Subnet, entering the Production Segment.
- The Cloud WAN Production Segment routing policy applies a send-to rule that directs the packet to the Inspection Segment, where the regional Inspection VPC NGFW is the destination for all internet-bound traffic. The NGFW in the Inspection VPC applies egress policy: FQDN-based URL filtering, application identification (App-ID where Palo Alto is deployed), and TLS inspection where enabled.
- Approved egress traffic is forwarded from the Firewall Subnet to the Public Subnet of the Inspection VPC, where the NAT Gateway resides and then forwards the traffic to the Internet Gateway destined to the internet.. All spoke VPC egress appears to originate from the Inspection VPC's NAT Gateway EIP pool, enabling centralized egress IP allow-listing at external partners.

East-West Traffic (Inter-Spoke):

- Traffic between VPC_Prod (us-east-1) and VPC_Prod (us-east-2), or between any two Production Segment spokes, is routed through the Cloud WAN backbone. Segment routing policy applies a send-via rule that directs inter-spoke traffic through the Inspection Segment NGFW before forwarding to the destination spoke — preventing any direct spoke-to-spoke path that would bypass inspection.

- Traffic between a Production Segment spoke and a DMZ Segment VPC follows the same enforcement model: Cloud WAN segment policy steers the packet through the Inspection Segment NGFW.

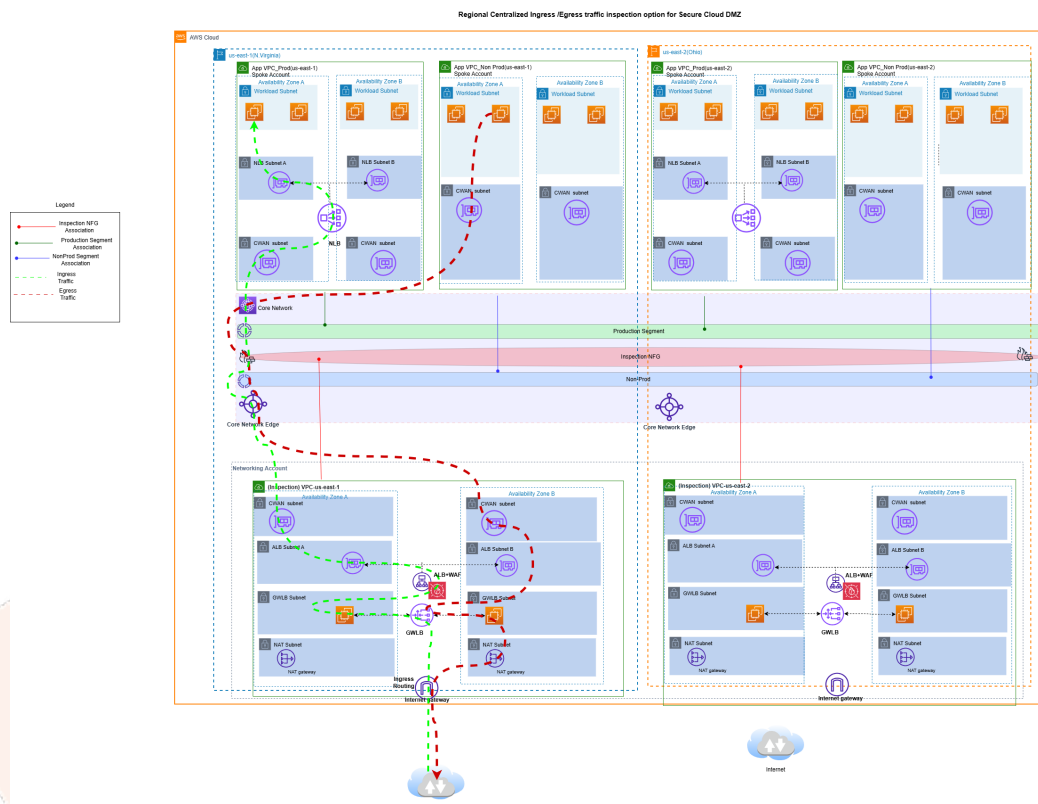
Pros:

- **Lower latency for inbound traffic** — requests enter directly at the workload VPC, minimizing hops
- **Higher availability** — no single point of failure for ingress; each workload VPC has independent internet entry
- **Reduced inter-region/inter-VPC data transfer costs** — inbound traffic doesn't traverse Cloud WAN or transit infrastructure
- **Simplified routing for ingress** — traffic stays local to the workload VPC after inspection
- **Better performance for latency-sensitive applications** — direct path from internet to application
- **Independent scaling** — each workload can scale ingress capacity independently
- **Native source IP preservation** — Gateway Load Balancer preserves the original client IP through the inspection flow
- **Simpler application logging** — applications see the true client IP without additional configuration

Cons:

- **Higher infrastructure costs** — requires IGW, Gateway Load Balancer endpoints, and firewall subnets per workload VPC
- **Increased operational complexity** — more endpoints and IGWs to manage and monitor
- **Inconsistent security posture risk** — distributed ingress points require careful governance to ensure uniform security policies
- **More public IP addresses required** — each workload VPC needs its own Elastic IPs for ingress
- **Harder to implement centralized logging and monitoring** — ingress logs scattered across multiple VPCs
- **Difficult to enforce global ingress policies** — changes must be replicated across all workload VPCs

5.1.3 Traffic Flows — Centralized Ingress & Egress Inspection



In this pattern, a single Inspection VPC per region serves as the exclusive internet boundary for both inbound and outbound traffic. The Inspection VPC contains the Internet Gateway, per-AZ NGFW endpoint cluster in the Firewall Subnet, a NAT Gateway in the Public Subnet, and Cloud WAN Attachment Subnets for backbone connectivity. Spoke VPCs — including both Production (VPC_Prod) and Non-Production (VPC_Non-Prod) — have no Internet Gateway and no public subnets. All internet traffic in both directions is steered by Cloud WAN segment routing policy through the Inspection VPC. The Cloud WAN core network enforces the Production Segment and Non-Prod Segment, both of which route exclusively through the Inspection Segment.

Inbound Internet Traffic (Centralized Ingress):

- An inbound internet request arrives at the Inspection VPC Internet Gateway — the only IGW in the region. No spoke VPC, workload VPC, or non-production VPC holds an IGW; all externally-initiated traffic enters the environment exclusively through the Inspection VPC.
- The IGW ingress route table directs the packet to the NGFW endpoint in the Inspection VPC Firewall Subnet for the matching AZ. The NGFW applies full stateful L4/L7 inspection, IPS, threat prevention, and (where Palo Alto is deployed) App-ID-

based policy. Traffic that does not match an explicit allow rule is dropped and logged before it can reach any spoke VPC.

- Approved inbound traffic exits the Firewall Subnet and is delivered to the ALB+WAF in the Inspection VPC Public Subnet. The ALB is the L7 termination point for all external-facing applications in the region — it is not located in the spoke VPC. The ALB terminates TLS, evaluates WAF rules against the HTTP/S request, and selects a backend target.
- The ALB forwards approved requests over the Cloud WAN backbone to an internal Network Load Balancer (NLB) residing in the Private Subnet of the destination spoke VPC (Production or Non-Prod Segment). The internal NLB is the sole ingress point into the spoke — it is not directly internet-reachable — and distributes traffic across the application backend: EC2 instances or Fargate ECS tasks in the same Private Subnet. This two-tier model (ALB+WAF in Inspection VPC internal NLB in spoke) decouples L7 termination and WAF enforcement from backend service discovery and scaling, and ensures no spoke VPC resource is directly addressable from outside the Cloud WAN backbone.
- Response traffic returns on the established session: EC2/ECS backend internal NLB (spoke VPC Private Subnet) Cloud WAN Attachment Subnet Inspection VPC ALB Firewall Subnet NGFW (stateful session match; return packets are permitted by existing flow state without re-evaluation of ingress rules) Inspection VPC IGW internet client. Response traffic does not traverse the centralized egress NAT Gateway path, which is reserved exclusively for new outbound sessions originated by spoke VPC workloads.

Outbound Internet Traffic (Centralized Egress):

- A spoke VPC (VPC_Prod or VPC_Non-Prod) initiates an outbound internet request. The subnet route table directs 0.0.0.0/0 to the Cloud WAN Attachment Subnet, entering the respective Production or Non-Prod Segment.
- The Cloud WAN segment routing policy applies a send-via rule that steers the packet to the Inspection Segment — directing it to the Inspection VPC NGFW in the same region. The NGFW applies egress policy: FQDN-based URL filtering, application control, and TLS inspection where enabled.
- Approved egress traffic is forwarded from the Firewall Subnet to the NAT Gateway in the Inspection VPC Public Subnet. The NAT Gateway performs source address translation, replacing the spoke VPC private IP with the Inspection VPC EIP, before forwarding to the Internet Gateway.
- The packet exits through the Inspection VPC Internet Gateway to the internet. All spoke VPC and non-production egress is sourced from the Inspection VPC NAT Gateway EIP pool, providing a single, controllable egress IP profile for external allow-listing.

East-West Traffic (Inter-Spoke and Cross-Segment):

- Traffic between Production Segment spoke VPCs (e.g., VPC_Prod us-east-1 to VPC_Prod us-east-2) traverses the Cloud WAN backbone. The Production Segment routing policy applies a send-via rule through the Inspection Segment, ensuring all inter-spoke traffic is inspected by the Inspection VPC NGFW before delivery to the destination spoke.
- Traffic between the Production Segment and the Non-Prod Segment is controlled by Cloud WAN segment isolation policy. Cross-segment communication (where permitted) is explicitly routed through the Inspection Segment NGFW, preventing any direct Production-to-Non-Prod path that would bypass inspection.
- This segment isolation model ensures that a compromise of a Non-Prod VPC cannot laterally reach Production workloads without traversing the centralized NGFW enforcement point.

Pros:

- **Simplified security management** — single ingress point for all inbound traffic with centralized policy enforcement
- **Easier compliance and auditing** — all ingress traffic flows through one inspection point
- **Lower infrastructure costs** — single IGW, fewer Gateway Load Balancer endpoints, consolidated firewall capacity
- **Centralized logging and monitoring** — unified visibility for all inbound traffic
- **Consistent security posture** — uniform policies applied at a single enforcement point
- **Easier to implement advanced security controls** — centralized DDoS protection, threat intelligence, and anomaly detection

Cons:

- **Higher latency** — additional hops through centralized ingress VPC before reaching workload
- **Increased data transfer costs** — all inbound traffic traverses Cloud WAN or transit infrastructure to reach workloads
- **Single point of failure risk** — centralized ingress VPC becomes critical; requires robust HA design
- **Potential bottleneck** — centralized ingress can become a performance constraint under high load
- **More complex routing** — requires Cloud WAN or Transit Gateway routing to distribute traffic to workloads
- **Blast radius concerns** — issues in centralized ingress affect all workloads

- **Source IP is lost or translated** — traffic traversing Cloud WAN from centralized ingress to workload VPCs undergoes NAT
- **Applications see internal VPC IPs** — the source IP becomes the centralized ingress VPC's NAT or firewall IP
- **Must rely on X-Forwarded-For headers** — ALB must insert client IP in HTTP headers; applications must parse them

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5.2 WAN Backbone Platform Design — Cloud WAN / Virtual WAN / NCC

The following describes the design pattern for each CSP's WAN backbone platform and how the Inspection Segment is implemented as a native construct within each core network.

5.2.1 AWS — Cloud WAN with Inspection Segment

AWS Cloud WAN provides a managed global network with a central Network Manager dashboard. The core network is divided into named Segments — the foundational routing construct. The design defines and not limited to following segments within the Cloud WAN core network:

- Inspection Segment — contains the NGFW (AWS Network Firewall or Palo Alto via GWLB), L7 ALB, WAF, and all DMZ-tier resources. All other segments route through this segment by Cloud WAN routing policy. Internet-facing VPCs attach to this segment.
- Production Spoke Segment — workload VPCs (application, data, shared services). Associated with spoke VPCs via Cloud WAN attachments. No direct route to other segments except via the Inspection Segment.
- Hybrid Segment — Direct Connect Gateway attachments and Site-to-Site VPN attachments connect to the Cloud WAN core network as Hybrid Segment attachments. This segment is isolated from the Production Spoke Segment. Cloud WAN segment routing policy steers all Hybrid Segment traffic through the Inspection Segment NGFW before any spoke is reachable. There is no Transit Gateway in this design.
- Partner/Extranet Segment — hosted Direct Connect connections and dedicated S2S VPN attachments for third-party and B2B partners. Isolated from both the Hybrid Segment and Production Spoke Segment. All partner-originated traffic traverses the Inspection Segment NGFW via segment routing policy before reaching any production spoke.

Cloud WAN Service Insertion is used to define the Inspection Segment NGFW as the service insertion endpoint. All segment-to-segment routes are configured with a send-via policy that forces traffic through the NGFW attachment before being forwarded to the destination segment. This is enforced at the core network routing layer not by static route tables in individual VPCs.

5.2.2 Azure — Virtual WAN with Inspection in Secured Virtual Hub

Azure Virtual WAN uses the Secured Virtual Hub concept a managed hub VNet with Azure Firewall Premium (or a supported 3rd-party NVA) embedded directly within the hub. The design pattern for Virtual WAN is:

- Secured Virtual Hub — Azure Firewall Premium (or Palo Alto via Gateway Load Balancer integration) is deployed inside the Virtual Hub. All VNet connections and ExpressRoute/VPN attachments to the hub have routing intent configured to steer all internet and private traffic through the hub firewall. This is the Inspection Segment equivalent.
- Spoke VNets — connected to the Secured Virtual Hub via Virtual WAN VNet connections. Routing intent on the hub is set to 'Send all traffic through Azure Firewall' no VNet has a direct path to another VNet or to the internet that bypasses the hub firewall.
- Hybrid Segment — ExpressRoute circuits and Site-to-Site VPN connections attach to the Secured Virtual Hub as branch/on-premises connections, logically forming the Hybrid Segment equivalent. Routing Intent steers all on-premises traffic through Azure Firewall Premium / NVA (Inspection Segment) before any spoke VNet is reachable.
- Partner / Extranet — isolated spoke VNets attached to the Virtual Hub with restricted routing intent. Partner traffic traverses the hub firewall; no peering to production spoke VNets.

Azure Virtual WAN Routing Intent is the key mechanism: it replaces manual route table configuration with a declarative policy that directs all private traffic and all internet traffic through the Secured Virtual Hub firewall — enforced at the VWAN platform layer.

5.2.3 GCP — NCC with Inspection via Network Virtual Appliance (NVA) Hub

GCP Network Connectivity Center (NCC) provides a hub-and-spoke routing model using Router Appliance and VPC spokes. The Inspection Segment is implemented as a dedicated NCC Hub VPC containing the NGFW NVA (Palo Alto or Cloud NGFW):

- NCC Hub — NCC Hub VPC (Inspection Segment) containing the NGFW NVA (deployed as a Managed Instance Group behind an Internal Passthrough Load Balancer for HA). The NCC Hub VPC is the Inspection Segment. All NCC spokes are configured to route traffic through the NCC Hub by default.
- NCC Spokes (VPC Spokes) — workload VPCs registered as NCC VPC spokes. Spoke-to-spoke traffic and spoke-to-internet traffic is routed via the NCC Hub (Inspection Segment) by NCC routing policy. No spoke has a direct peering route to another spoke outside the NCC hub.
- Hybrid Segment — Cloud Interconnect and Cloud VPN attachments are registered as NCC hybrid spokes in a dedicated Hybrid Segment. These are isolated from

production VPC spokes by NCC routing policy. On-premises traffic enters via the Hybrid Segment and is steered through the NCC Hub NGFW (Inspection Segment) before any production VPC spoke is reachable.

- Partner Spokes — dedicated NCC VPC spokes for partner and B2B traffic, isolated from production VPC spokes. All partner traffic transits the NCC Hub NGFW.

GCP Cloud NGFW (Palo Alto native) can also be deployed in Zonal or Regional mode within the NCC Hub VPC (Inspection Segment), providing cloud-native firewall policy enforced at the NCC Hub level. NCC custom route advertising ensures all inter-spoke and internet-bound traffic is steered through the NCC Hub NGFW before egress or spoke delivery.

5.2.4 Cross-CSP Inspection Segment Consistency

Across all three CSPs, the Inspection Segment design applies a consistent set of principles:

Design Principle	AWS Cloud WAN	Azure Virtual WAN	GCP NCC
Inspection Segment construct	Named Segment in Cloud WAN core network	Secured Virtual Hub with Azure Firewall Premium / NVA	NCC Hub VPC with NGFW NVA
NGFW placement	NGFW attachment within Inspection Segment (AWS Network Firewall / Palo Alto via GWLB)	Azure Firewall Premium or Palo Alto NVA embedded in Secured Virtual Hub	Palo Alto / Cloud NGFW in NCC Hub VPC behind ILB
Traffic steering mechanism	Cloud WAN Service Insertion — send-via policy per segment	Virtual WAN Routing Intent — all private + internet traffic through hub firewall	NCC custom route advertising; ILB as next hop for all inter-spoke routes
Spoke attachment model	VPC attachments to Cloud WAN core network; segment association per VPC	VNet connections to Secured Virtual Hub	NCC VPC spokes registered to NCC Hub

On-premises attachment	Direct Connect Gateway attachment to Cloud WAN core network	ExpressRoute / VPN to Secured Virtual Hub	Cloud Interconnect / VPN as NCC hybrid spokes
Internet ingress/egress	Inspection VPC with IGW attached to Inspection Segment; traffic steered via service insertion	Internet traffic routed through Secured Virtual Hub Azure Firewall via routing intent	Cloud NAT / IGW in NCC Hub VPC (Inspection Segment); traffic via ILB next-hop to NGFW
Partner / extranet isolation	Dedicated Partner Segment in Cloud WAN core network	Isolated spoke VNet with restricted hub routing intent	Dedicated NCC VPC spoke with restricted route advertisement
East-west inspection	Segment routing policy prevents spoke-to-spoke direct path; all via Inspection Segment	Routing intent directs all private traffic through hub firewall	NCC Hub as mandatory next-hop; no spoke-to-spoke peering

6.0 How the Proposed Architecture Addresses Each Requirement

This section provides explicit traceability between every governance and design requirement and the specific architectural mechanism in the proposed Cloud WAN / Virtual WAN / NCC backbone with Inspection Segment design. For each requirement, the architectural response describes the exact WAN platform component, its position in the traffic path, and the enforcement mechanism.

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6.1 On-Premises Sandwich Model and Proxy Requirements

The proposed Cloud WAN / Virtual WAN / NCC architecture maps the on-premises sandwich model to WAN-native constructs. Critically: in the cloud design, both the external (internet-facing) and internal (spoke-facing) firewall functions are performed by a cloud-native firewall or vNGFW within the Inspection. The WAN routing fabric enforces that all traffic transits the Inspection Segment; the NGFW applies separate ingress and egress policy rule sets for each direction.

Requirement	Cloud WAN / VWAN / NCC Architecture Component	How It Is Addressed
External Firewall Function — internet-facing enforcement boundary (on-premise: Outer Firewall)	NGFW within the Inspection Segment (cloud-native: AWS Network Firewall / Azure Firewall Premium / Cloud NGFW, or 3rd-party: Palo Alto / Fortinet via GWLB or ILB) performing both external and internal firewall functions	Internet traffic arrives at the inspection VPC Internet Gateway where the NGFW/Cloud-native firewall is the first and only enforcement boundary. Cloud WAN handles all inter-segment routing natively. The NGFW of Cloud Native firewall applies full stateful L4–L7 inspection, IPS signatures, and threat prevention for both north-south and east-west traffic
DMZ / Inspection Zone — isolated enforcement tier hosting NGFW, ALB, WAF, and	Inspection VPC / Hub VPC contains the NGFW, L7 Load Balancer, WAF, and all DMZ-tier resources (public-facing subnets).	The Inspection Segment VPC is the cloud DMZ — it hosts the NGFW, the L7 ALB, the WAF, and all DMZ-tier resources. Cloud WAN segment routing policy ensures all traffic entering the WAN core network is steered through the Inspection Segment. The NGFW handles both ingress enforcement (outer firewall function) and egress enforcement toward spoke VPCs (inner firewall function) through separate policy rule sets on the same engine.

internet-facing services		
Internal Firewall Function spoke-facing enforcement boundary (on-premises: Inner Firewall)	AWS: Cloud WAN segment routing policy on Inspection Segment egress + spoke VPC Security Groups Azure: VWAN Routing Intent private traffic policy + spoke VNet NSGs GCP: NCC Hub ILB next-hop routing + spoke VPC Firewall Rules	The same NGFW engine in the Inspection Segment performs the internal (inner) firewall function through a distinct egress policy rule set. Internet-facing policy (external function) and spoke-facing policy (internal function) are separate rule sets on policy engine of firewall devices. Cloud WAN / VWAN / NCC segment routing policy enforces that all spoke-bound traffic exits via the Inspection Segment NGFW egress policy.
Proxy / Termination Layer — mandatory between internet zone and internal spokes; no unmediated path (Section 2.4)	L7 ALB (AWS ALB / Azure Application Gateway / GCP HTTPS LB) with WAF, AWS Network Firewall Proxy provisioned within the Inspection Segment of Cloud WAN / Secured Virtual Hub / NCC Hub VPC	The ALB acts as the mandatory proxy. All inbound connections terminate at the ALB; the client TLS session ends here. The ALB opens a new connection to the internal application target. This means no TCP session traverses from the internet to an internal application server. The WAF inspects all HTTP/HTTPS payloads at L7 before forwarding. For vpc outbound connection Network firewall proxy inspect and filter vpc outbound connections to prevent data exfiltration and malware intrusion. This fully satisfies the proxy /termination layer governance requirement.
Hybrid Segment — VPN / Direct Connect	AWS: DirectConnect, Connect and Site-to-Site VPN attach to the Hybrid Segment in Cloud WAN core network Azure:	VPN and Direct Connect attach to the WAN backbone as Hybrid Segment attachment. Authenticated VPN sessions and on-premises traffic enter the WAN fabric at the Hybrid Segment. Cloud WAN / VWAN / NCC segment routing policy then steers all Hybrid Segment

connectivity (on-premise s: DMZ sandwich for client connectivity)	ExpressRoute and VPN terminate on the Secured Virtual Hub (Hybrid Segment equivalent) GCP: Cloud VPN registered as NCC hybrid spoke in the Hybrid Segment	traffic through the Inspection Segment NGFW before any spoke VPC/VNet is reachable. The Hybrid Segment is shared with Production Spoke Segments and other spokes for connectivity to on premise workload, however traffic is steered toward the inspection segment before reaching on-prem.
Partner / M&A, B2B Segment third-party connectivity (on-premise s: DMZ sandwich for partner traffic)	AWS: Partner/M&A and B2B Segment in Cloud WAN core network (hosted Direct Connect + S2S VPN attachments) Azure: Isolated partner spoke VNet on Secured Virtual Hub with restricted routing intent GCP: Dedicated partner NCC VPC spoke in the Partner Segment with restricted route advertisement	All partner circuits (hosted Direct Connect connections, dedicated S2S VPN) attach to the Partner/Extranet Segment — isolated from the Hybrid Segment, Inspection Segment, and Production Spoke Segment. Cloud WAN / VWAN / NCC segment routing policy routes all partner-originated traffic through the Inspection Segment NGFW before reaching any production spoke. Partners have zero routing visibility to production or hybrid segments

6.2 GCS Section 3 — Port and Service Access Control

GCS requires that network access is limited to authorized services and ports, all unapproved flows remain blocked, and all access to restricted environments is documented with business justification. The Cloud WAN / Virtual WAN / NCC architecture enforces this at four independent layers — with WAN segment routing policy as the outermost layer and host-based controls as the innermost.

GCS Directive	Architectural Enforcement Layer	Specific Mechanism
Network access limited to authorized services and ports only	Layer 1: Cloud WAN / VWAN / NCC segment routing policy Layer 2: Inspection Segment NGFW ingress policy (external function) Layer 3: Inspection Segment NGFW egress policy (internal function) Layer 4: Spoke Security Groups / NSGs + Host-based controls	The WAN segment routing policy is the outermost enforcement layer — it defines which segments can communicate and forces all inter-segment traffic through the Inspection Segment. The Inspection Segment NGFW maintains an explicit allow list of approved port/service combinations. Security Groups / NSGs on all spoke resources follow deny-all default. Each rule is tagged with the PTx ticket reference for audit traceability.
Inbound and outbound unapproved flows must remain blocked	Default-deny at WAN segment routing policy + Inspection Segment NGFW (ingress and egress policies) + spoke Security Groups / NSGs + VPC NACLs as secondary stateless layer	Cloud WAN / VWAN / NCC segment routing policy has no default route between segments — spoke-to-spoke and internet-to-spoke routes are only created explicitly through the Inspection Segment. The Inspection Segment NGFW has a default deny-all at the bottom of both ingress and egress policies. An unapproved flow is blocked at the WAN routing layer before it even reaches the NGFW, providing defense-in-depth.
Sensitive data secured via approved	L7 ALB TLS termination within Inspection Segment + mTLS for backend connections + WAN backbone encryption	The ALB in the Inspection Segment enforces minimum TLS 1.2 (TLS 1.3 preferred). HTTP listeners are 301-redirect only. Backend connections to spoke VPCs use HTTPS with certificate validation. B2B partner listeners require mTLS. With AWS VPC encryption

encryption on all communication channels		control, encryption is enabled in transit within and across VPCs; Azure VWAN backbone is Microsoft-encrypted; GCP NCC uses MACsec /encrypted interconnect where supported.
Well-known ports not exposed outside enterprise control without approved business reason	Inspection Segment NGFW outbound policy + Cloud WAN / VWAN / NCC segment egress routing policy + spoke Security Group / NSG egress rules	All outbound internet traffic routes through the Inspection Segment NGFW by CloudWAN policy, NGFW/Cloud-Native rule change controls all outbound port exposure across every spoke in the WAN. Spoke Security Groups / NSGs restrict local egress to only required destination ports. Any exception requires PTx-documented approval before an NGFW rule or segment routing policy change is committed.

6.3 GCS Sections 4 & 5 — Non-Standard Ports and Port Forwarding

Specific mechanisms:

- AWS: No NAT instance port-forwarding rules; ALB listener rules use path-based and host-based routing only. Any non-standard Security Group rule requires a tagged Change ticket attached to the Terraform plan before pipeline approval gate passes.
- Azure: Application Gateway uses URL-rewrite rules and path-based routing. No Azure NAT rules are used for port forwarding to application servers. NSG rules are deployed via Azure Policy with a required justification tag.
- GCP: HTTPS Load Balancer uses URL maps for host and path-based routing. VPC Firewall rules with allow actions for non-standard ports require an Org Policy exception workflow approval.

6.4 GCS Section 6 — Insecure Services Prohibition

The proposed architecture blocks all GCS-prohibited insecure services at multiple independent enforcement points. The following details how each prohibited service is addressed by a specific architectural mechanism.

In se cu re Se rv ice	P o rt	Architecture Enforcement Mechanism
FTP	tcp / 21	Explicit DENY in Inspection Segment NGFW ingress policy. Security Group on all DMZ-tier resources has no inbound rule for tcp/21. Because all WAN traffic is steered through the Inspection Segment by Cloud WAN / VWAN / NCC routing policy, this single NGFW deny rule covers all ingress paths — no separate per-VPC rule needed.
TF TP	udp / 69	Explicit DENY in Inspection Segment NGFW UDP ingress policy enforced for all WAN-attached VPCs/VNets. No Security Group inbound rule permits udp/69 at any tier. NACL explicit deny as secondary stateless control.
H TTP	tcp / 80	HTTP is blocked at the application target layer — no target group accepts HTTP from outside the ALB. The ALB listener on port 80 is configured exclusively for permanent 301 redirect to HTTPS (port 443). The WAF associated with the ALB has a managed rule blocking direct HTTP access to application paths. Security Groups on application servers deny tcp/80 from any source other than the ALB.
L D A P (u n e nc	tcp / 389	Security Groups / NSGs on all directory service resources deny tcp/389 from any source outside the trusted spoke zone. Inspection Segment NGFW egress policy explicitly denies tcp/389 toward any destination outside the trusted internal segment.

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Te In et	t c p / 23	Inspection Segment NGFW ingress and egress policies include explicit DENY for tcp/23 enforced for all WAN-attached segments. All spoke Security Groups / NSGs have no inbound rule for tcp/23. Host-based firewall (iptables / Windows Firewall) blocks telnet as defense-in-depth.
N et BI O S / S M B v1 - v2	t c p - u d p / 1 3 7 - 4 45	All Security Groups / NSGs deny NetBIOS ports (137-139) globally. SMB on tcp /445 is permitted only within trusted internal spoke VPC subnets between pre-approved resources using SMB v3+. SMB is not permitted to traverse the Inspection Segment boundary (the WAN segment routing policy has no route for NetBIOS/SMB between segments).
S N M P v1 /v2	u d p / 1 6 1 - 1 62	Inspection Segment NGFW ingress policy explicitly denies SNMP v1/v2 (udp /161-162) from any external source — applied to all WAN segments. Security Groups / NSGs on monitoring targets permit udp/161 only from the dedicated NMS server IP using SNMPv3.